Efficient Solar Filament Segmentation at Ensemble Parity

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

Solar filaments are thin, low-contrast chromospheric structures whose instance segmentation is dominated by annotator disagreement and by fragmentation or over-merging errors. We describe a compact ConvNeXt-atto U-Net (4.28 M parameters) trained with knowledge distillation, randomized-annotator targets and stochastic weight averaging, then deployed with single-pass 2048-pixel inference, eight-view dihedral test-time augmentation and a decoupled "KEEP" instance-acceptance rule. On the 107-file validation split the student reaches an exact multi-annotator Panoptic Quality (PQ) of 0.4591 and a public leaderboard score of 0.38—equal to a 171 M four-model ensemble—while using 107 GMAC and 103 ms per image (40× fewer parameters and ∼30× fewer MACs at equal leaderboard score). Under the official IoU> 0.5 matching, the submitted system produces no one-to-many or many-to-one relations; at IoU> 0 only 2.4% of ground-truth instances and 5.1% of predictions have multiple partners. The decoupled KEEP rule alone is worth +0.0158 nested-CV PQ and moved the leaderboard from 0.35 to 0.37 with no retraining; full-disk inference and TTA added the final quantum to 0.38.

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

Our contribution is threefold. (i) We show that the binding constraint of the previous system was not model capacity but a coupled post-processing rule: decoupling the contour threshold from instance acceptance yields +0.0158 nested-CV PQ and +0.02 leaderboard without retraining. (ii) A 4.28 M-parameter distilled student, fine-tuned with randomized-annotator targets and SWA, matches a 171 M four-model ensemble at PQ 0.38. (iii) The resulting pipeline is single-pass, deterministic and reproducible; its weights, inference notebook and verification scripts are released.

2 Methodology

Data and split. MAGFiLO contains 707 GONG H-α full-disk images (2048 × 2048) with 1–3 expert annotators per image (1,154 records; 8,199 instance masks). We split *files*, not annotator records, 85/15 (600 train / 107 validation), so no image appears in both sets through a different annotator. The exact metric is computed against every annotator's mask with all-pairs IoU> 0.5 matching.

Preprocessing. Images are read as raw 8-bit grayscale. At inference the solar disk is detected on a 4× downsample (threshold, morphological close, convex hull, erosion) and used only as an outside-disk gate. The single-channel image is replicated to three channels and normalized on the GPU with ImageNet mean/std, matching the pretrained stem. No CLAHE or radial flattening is used; both were falsified on this data.

Model and training. The student is a ConvNeXt-atto encoder [5] with a four-stage U-Net decoder [6] (channels 128/64/32/16), 4.28 M parameters. It is trained on 1024-pixel crops with a balanced loss 0.5 BCE + 0.5 Tversky($\alpha = \beta = 0.5$), AdamW (lr 10^{-4}), linear warmup followed by cosine annealing, and per-batch EMA (raw weights are used for inference; EMA loses instance PQ). Targets are the first annotator's mask; a six-epoch recall fine-tune switches to *random_annotator* (all 1,154 records), starts from the complete-GT checkpoint, uses knowledge distillation from a five-teacher ensemble [3], and averages the last three epochs (SWA). The fine-tune selects checkpoints by full-image PQ under the deployed rule, not by pixel Dice.

Inference and post-processing. The network is fully convolutional, so the whole 2048 × 2048 image is processed in one forward pass (107 GMAC, 103 ms FP16), optionally with eight dihedral views (D4 TTA8) whose sigmoid maps are averaged. We then apply the decoupled KEEP rule: form 8-connected components of $p > t_{\text{lo}}$, and accept a component iff

$$\text{peak}(p) > t_{\text{hi}}, \qquad \text{mean}(p) \geq m_{\text{p}}, \qquad \text{area} \geq a_{\text{min}}.$$

The deployed student uses $t_{\text{lo}} = 0.35$, $t_{\text{hi}} = 0.99$, $m_{\text{p}} = 0.85$, $a_{\text{min}} = 100$. Components are exported as COCO RLE. The nested-CV optimum was $a_{\text{min}} = 300$; the two are indistinguishable on validation (0.45909 vs 0.45906), and the submitted CSV uses 100. The accuracy reference (S2) uses a 171 M four-model 1024-pixel ensemble with $t_{\text{lo}} = 0.35$, $t_{\text{hi}} = 0$, $m_{\text{p}} = 0.80$, $a_{\text{min}} = 100$; the leanest point (S4) uses a 64-wide decoder and $t_{\text{lo}} = 0.30$, $t_{\text{hi}} = 0.99$, $m_{\text{p}} = 0.85$, $a_{\text{min}} = 100$.

3 Evaluation

Protocol. We use the exact competition metric: multi-annotator per-image ground truth and all-pairs IoU> 0.5 matching, giving $\text{PQ} = \frac{\sum_{(p,g)\in TP} \text{IoU}(p,g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}$ [4]. Rule and ensemble choices are selected with nested 5-fold cross-validation on the 107 validation files and compared with a paired file-level bootstrap (4,000 resamples).

Results. Table 1 summarises the three operating points. The submitted student reaches 0.4591 validation PQ and 0.38 on the public leaderboard, equal to the 171 M ensemble, at 40× fewer parameters and about 30× fewer MACs on the same single-pass path. The 3.75 M leanest point is 0.0008 better in validation but lands one leaderboard quantum lower (0.37), which is within the public-subset

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

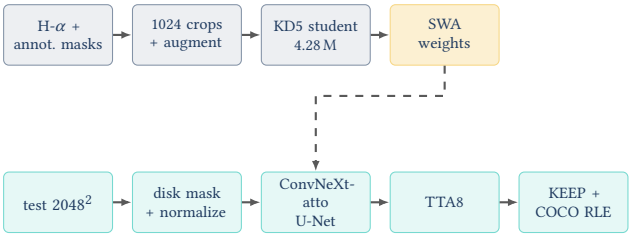


Figure 1: End-to-end pipeline. Navy: training (once), with KD5 teachers and SWA. Teal: inference per test image; the gold SWA weights initialize the U-Net. The released notebook reproduces the submission from the weights without retraining.

System	Params	MACs	Val PQ	LB
S2 ensemble (4×1024, TTA8)	171.0 M	3.23 TMAC	**0.4773**	**0.38**
S3 student (submitted)	**4.28 M**	**107 GMAC**	0.4591	**0.38**
S4 student (64-wide, leanest)	3.75 M	64.8 GMAC	0.4599	0.37

Table 1: Validation PQ (107 files, exact metric) and public leaderboard. MACs are one forward per model, not multiplied by the eight TTA views.



(a) IoU **(b) Dice**

(c) Fragmentation / over-merging relations

Figure 2: Matched-pair IoU/Dice and one-to-many/many-to-one relations for S3 on the 107 validation files. Relations are shown at IoU>0 (left bars) and under the official IoU>0.5 matching (right bars).

Effect (nested OOF)	ΔPQ	95% CI
KEEP vs coupled	+0.0158	[0.0087, 0.0231]
tiny + randann	+0.0127	[0.0045, 0.0217]
tiny + randann + SWA	+0.0105	[0.0032, 0.0177]
dec128 + randann + SWA	+0.0051	[−0.0056, 0.0167]

Table 2: Paired file-level bootstrap (4,000 resamples) for the rule and fine-tune deltas. All intervals are nested 5-fold OOF.

noise floor. The historical 0.35 plateau was therefore a decision-rule and inference-path problem, not a model-capacity ceiling.

Accuracy distributions. On the 933 matched validation pairs, mean IoU is 0.689 (median 0.695) and mean Dice is 0.813 (median 0.820); Fig. 2 shows the distributions. The matched-pair mass is concentrated at high overlap, consistent with a contour that preserves thin barbs rather than smoothing them.

Fragmentation and over-merging. Under the official IoU>0.5 matching, the submitted system has *zero* one-to-many and many-to-one relations: every matched ground-truth instance has exactly one prediction and vice versa. At the looser IoU>0 diagnostic, 2.4% of ground-truth instances overlap more than one prediction and 5.1% of predictions overlap more than one ground-truth instance (Fig. 2, right). The KEEP rule is what removes the residual fragmented/merged candidates before they reach the metric.

Error anatomy. At the deployed rule S3 has 933 matched instances, 474 false positives and 461 false negatives. The FNs decompose into 292 missed (63.3%), 163 boundary (35.4%) and 6 merged (1.3%). Moving from the old coupled rule to KEEP on the same TTA8 probabilities changes TP/FP/FN from 916/575/478 to 933/474/461, raising RQ from 0.635 to 0.666 while SQ moves from 0.686 to 0.689: the rule adds 17 true positives and removes 101 false positives. Post-acceptance dilation and erosion are both negative (dil1 −0.025, dil2 −0.094, ero1 −0.022 PQ), so the residual is not a thin-boundary problem.

Strengths and weaknesses. The system is fast, small, and free of explicit fragmentation/merging under the official matching. Its weakness is recall: 63.3% of remaining FNs are pure misses and 35.4% are boundary/agreement failures, and the labels themselves are noisy (inter-annotator union-mask IoU 0.467), so validation absolute levels are optimistic relative to the sparser test labels. The public leaderboard is quantised at 0.01 and cannot arbitrate sub-0.01 differences; we therefore report deltas and bootstrap intervals, not just absolute PQ. More positive training data (5-fold over all 707 images) is the most promising remaining lever.

Qualitative comparison. Fig. 3 shows zoomed validation examples. The prediction follows the elongated body and barbs in (a), the curved limb filament in (b), and one annotator's drawing in (d); the main failure in (c) is a missed low-contrast filament, not a spurious or merged one. Limb artifacts are suppressed by the solar-disk gate.

Uncertainty and efficiency. Table 2 makes the uncertainty explicit: the KEEP rule and the tiny recall fine-tune are significant; the dec128 fine-tune is not. Table 1 gives the efficiency: S3 is about 30× left of S2 at the same 0.38 LB, and S4 is the leanest point but one LB quantum lower.

Reproducibility. The released repository contains the submitted weight (16.4 MB), a requirements file, a notebook that reproduces the CSV from the test images without retraining, and a 13-check submission verifier. The submitted CSV has 1,279 rows over 177/180 test images (the three empty images are legitimate) and passes all

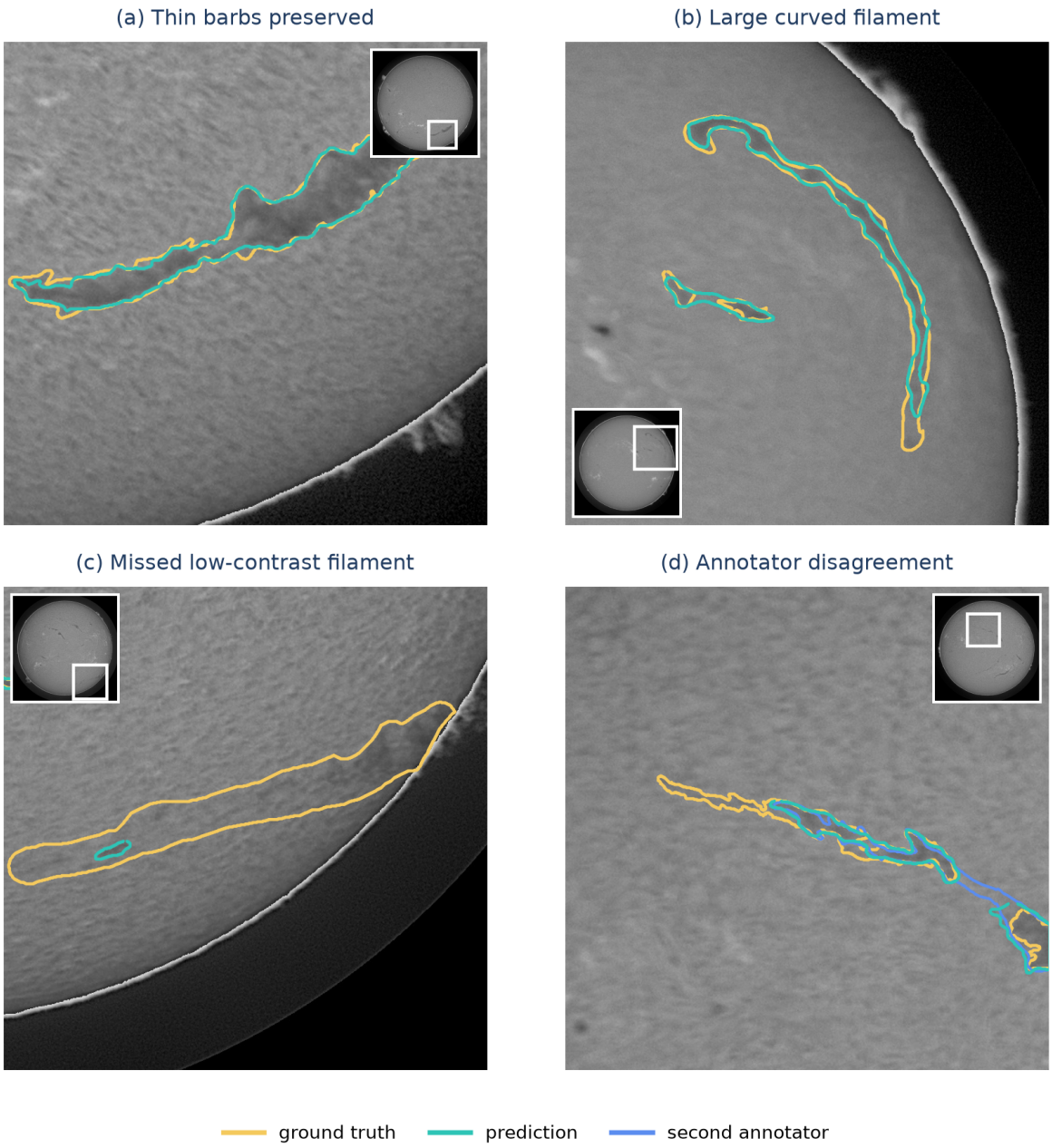


Figure 3: Zoomed validation examples. Gold: ground truth; teal: prediction; blue in (d): a second annotator. (a) thin barbs, (b) a large curved filament near the limb, (c) a missed low-contrast filament, (d) annotators disagree at the ends while the prediction follows one. Insets show the full-disk location.

checks. Recomputing from the released weight yields 1,280 rows: one boundary component on one image differs because the reference CSV was written from the FP16 probability cache, not re-inferred.

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